WELDING AND MATERIAL JOINING PROCESSES - PART I

Fundamentals • Pressure Welding • Gas Welding • Electric Arc Welding

Reference: Elements of Workshop Technology, Vol. II - Hajra Choudhury

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1. Definition of Welding and Weldability

Welding is a permanent joining process in which two or more parts are united by the application of heat, pressure, or a combination of heat and pressure, with or without the addition of filler metal. The joint is produced by establishing metallurgical bonding across the interface.

1.1 Weldability

Weldability is the ability of a material to be welded under given fabrication conditions into a specific, suitably designed structure and to perform satisfactorily in service. It is not only a property of the metal; it also depends on the welding process, joint design, thickness, restraint, filler metal, heat input and required service conditions.

- Good weldability means sound joints can be obtained without cracking, excessive porosity, unacceptable distortion or serious loss of mechanical properties.
- Low-carbon steels usually possess good weldability. High-carbon steels and some alloy steels require preheating, controlled heat input and post-weld heat treatment.
- Aluminium, copper, cast iron and stainless steels can be welded successfully, but each needs suitable process control, filler metal and shielding.

1.2 Factors affecting weldability

Factor	Effect on welding
Chemical composition	Controls hardenability, hot cracking, oxidation and metallurgical transformations. Carbon equivalent is important for steels.
Thermal conductivity	High conductivity carries heat away rapidly and may require high heat input or preheating.
Thermal expansion	Large expansion/contraction increases distortion and residual stresses.
Melting temperature / range	Influences heat source selection and solidification behaviour.

Factor	Effect on welding
Surface oxide or contamination	May prevent fusion or cause inclusions and porosity unless removed or fluxed.
Thickness and restraint	Thick or highly restrained joints cool rapidly and develop greater residual stresses.
Service requirement	Fatigue, corrosion, pressure, temperature and impact requirements determine acceptable procedure.

2. Classification of Welding Processes

Welding processes may be broadly classified according to whether the base metal is melted during joining.

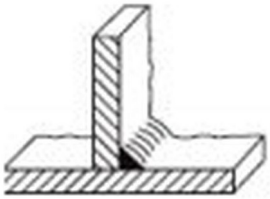
Main class	Principle	Examples
Pressure / solid-state or resistance-assisted welding	Joint is obtained mainly by pressure; base metal may remain solid or only local interfacial melting occurs.	Forge welding, resistance butt, spot, seam, projection, percussion, ultrasonic welding.
Fusion welding	The faying surfaces are melted to form a weld pool; filler metal may or may not be added.	Gas welding, metal arc, TIG, MIG, submerged arc, plasma arc, thermit, electron beam, laser beam.
Allied joining processes	Base metals usually do not melt; a lower-melting filler metal flows into the joint.	Soldering and brazing.

3. Welding Positions and Types of Welding Joints

The principal positions are flat (down-hand), horizontal, vertical and overhead. Flat welding is easiest because gravity helps retain the molten pool. Vertical and overhead welding require smaller pools, careful electrode manipulation and suitable current to prevent metal sagging.

3.1 Types of welded joints and weld forms

Form of weld	Illustration	BS symbol
Butt weld between flanged plates (the flanges being melted down completely)		
Square butt weld		
Single-V butt weld		
Single-bevel butt weld		

Form of weld	Illustration	BS symbol
Single-V butt weld with broad root face		
Single-bevel butt weld with broad root face		
Single-U butt weld		
Single-J butt weld		
Backing or sealing run		
Fillet weld		
Plug weld (circular or elongated hole, completely filled)		

4. Pressure Welding Processes

4.1 Forge welding

Forge welding is one of the oldest welding processes. The surfaces to be joined are cleaned, heated in a forge to a plastic or near-welding temperature and then united by hammering or pressing. A flux such as borax may be used to dissolve oxides and prevent further oxidation.

1. Prepare and scarf the surfaces so oxide can be expelled and adequate contact obtained.
2. Heat the parts uniformly to the required welding temperature.
3. Apply flux where needed and bring the surfaces together.
4. Hammer or press from the centre outward to expel flux/scale and complete the bond.

- Advantages: simple equipment, no electrical source required, useful for wrought iron and traditional smith work.
- Limitations: operator-skill dependent, poor suitability for large production, difficult temperature control, limited material range.

4.2 Concept of resistance welding

Resistance welding develops heat at the joint due to electrical resistance when a very high current passes through contacting surfaces for a short time. Pressure is simultaneously applied by electrodes. The approximate heat generated is $Q = I^2Rt$, where I is current, R is effective resistance and t is current time.

Process	Working principle	Typical application
Resistance butt welding	Ends of bars/wires are clamped, brought into contact, heated by resistance and upset under axial pressure.	Rods, wires, rails and simple sections.
Spot welding	Overlapped sheets are squeezed between copper electrodes; current forms a local weld nugget.	Automobile bodies, cabinets, sheet assemblies.
Seam welding	Rotating wheel electrodes make overlapping nuggets or a continuous seam.	Fuel tanks, drums, leak-tight sheet joints.
Projection welding	Embossed projections localize current and pressure; several welds can be made in one stroke.	Nuts, studs, cross-wire joints, brackets.
Percussion welding	Parts are brought together rapidly while a short high-energy electrical discharge creates local heat; impact pressure completes the weld.	Fine wires, dissimilar metals, small electrical components.

4.3 Advantages and limitations of resistance welding

- High production rate and easy automation; no flux or filler normally required.
- Small heat-affected zone and relatively low distortion for sheet metal.
- Clean process suitable for mass production.
- Equipment is comparatively expensive and electrical demand is high.
- Joint design and access are restricted by electrode arrangement; surface condition affects consistency.

5. Fusion Welding - Gas Welding

5.1 Oxy-acetylene welding: concept

Oxy-acetylene welding uses the heat of combustion of acetylene in oxygen. The flame temperature is sufficiently high to melt common engineering metals. The edges of the work are fused and, when necessary, a filler rod of suitable composition is added. Flux is used for metals whose oxides cannot be controlled by the flame alone.

5.2 Welding equipment

- Oxygen cylinder and acetylene cylinder, each fitted with valves and pressure regulators.
- Pressure gauges, colour-coded hoses and non-return/flashback protection devices.
- Welding torch/blowpipe with mixing chamber and interchangeable nozzles.
- Filler rods and fluxes appropriate to the base material.
- Spark lighter, goggles, gloves, apron and other safety equipment.

5.3 Types of gas flames

Flame	Gas condition	Characteristics and use
Neutral	Approximately balanced oxygen and acetylene	Clearly defined inner cone; general welding of mild steel, stainless steel, cast iron, copper and many non-ferrous metals.

Flame	Gas condition	Characteristics and use
Carburizing / reducing	Excess acetylene	Has an acetylene feather; reducing atmosphere. Used where oxidation must be minimized and for selected alloys/hardfacing.
Oxidizing	Excess oxygen	Short sharp inner cone and hissing sound; used for brass/bronze and special operations. Excess oxygen can oxidize steel.

5.4 Gas-welding techniques

Two common techniques are leftward (forehand) and rightward (backhand) welding.

Technique	Movement	General features
Leftward / forehand	Torch points in the direction of travel; filler rod precedes the flame.	Common for thin sheets; easy observation of joint edges, wider bead, lower penetration.
Rightward / backhand	Torch points back toward the completed weld while travelling forward.	Often used for thicker plate; better heat utilization, deeper penetration and narrower bead.

5.5 Advantages, disadvantages and applications

- Advantages: portable, simple, inexpensive compared with many power welding systems, useful for repair, thin sheet and maintenance.
- Disadvantages: lower speed, larger heat-affected zone and distortion, slower for thick sections, operator skill strongly affects quality.
- Applications: sheet-metal fabrication, tubes, repair work, maintenance, non-ferrous welding, brazing, heating and cutting.

6. Electric Arc Welding

6.1 Principle of metal arc welding

An electric arc is established between an electrode and the workpiece. The arc converts electrical energy into intense heat, melting the joint edges and, when the electrode is consumable, the electrode tip. The molten metal solidifies to form the weld. Shielding is provided by flux, gas or both, depending on the process.

6.2 Metal arc welding equipment

- AC transformer or DC generator/rectifier/inverter power source.
- Electrode holder, welding cables, work/earth clamp and connectors.
- Electrodes, chipping hammer, wire brush and joint-preparation tools.
- Helmet/hand shield with correct filter shade, gloves, apron and ventilation arrangements.

6.3 AC and DC metal arc welding - comparison

Point	AC welding	DC welding
Power source	Transformer; simple and economical.	Generator, rectifier or inverter; usually costlier.
Polarity	Alternates every half-cycle; no fixed polarity.	Can select DCEN or DCEP.
Arc starting/stability	May be less stable with some electrodes, especially at low current.	Generally smooth and stable; good for difficult electrodes.

Point	AC welding	DC welding
Arc blow	Much less troublesome.	Can suffer magnetic arc blow.
Electrode selection	Requires electrodes suitable for AC.	Wide electrode choice and polarity control.
Special use	Good general fabrication and long leads.	Useful for thin sheet, positional welding and controlled penetration.

6.4 Arc voltage and welding current

Arc voltage is mainly related to arc length: a longer arc tends to increase voltage and produce a wider, less concentrated heat source. Welding current primarily controls melting rate and penetration. Too low a current causes poor fusion and unstable arc; too high a current causes excessive spatter, undercut, burn-through and electrode overheating. Correct values depend on electrode diameter, type, position, joint thickness and polarity.

6.5 Polarity

DCEN means electrode negative and work positive; DCEP means electrode positive and work negative. The practical effects on penetration, electrode melting rate and oxide-cleaning action depend on the particular electrode and arc process. Electrode manufacturers specify the recommended polarity.

6.6 Arc blow

Arc blow is the unwanted deflection of a DC welding arc by magnetic fields around the welding circuit. It is most noticeable near corners, ends of joints and high-current welds.

- Effects: irregular bead, undercut, lack of fusion, spatter and difficulty directing the molten metal.
- Remedies: use AC where possible, reduce current/arc length, change earth-clamp position, weld toward the earth connection, use back-step sequence or magnetic balancing.

6.7 Electrodes: types and specification

Type	Description
Bare electrode	Uncoated metal rod/wire; little shielding, therefore limited use for ordinary steel welding.
Lightly coated electrode	Thin coating improves arc stability and provides some shielding/deoxidation.
Heavy-coated / shielded electrode	Flux coating produces gases and slag, stabilizes arc, adds deoxidizers/alloying ingredients and controls bead shape.
Non-consumable electrode	Carbon/graphite or tungsten; does not intentionally become filler metal.

A common covered-electrode identification system uses a classification such as E6013: E = electrode; 60 = minimum tensile strength of deposited weld metal in ksi (approximately 60 ksi); 1 = all-position capability; the last digits indicate coating/current characteristics. Exact meaning should be read from the applicable electrode standard and manufacturer data.

6.8 Angularity of electrodes

- Work angle: angle of the electrode across the joint; selected to distribute heat correctly between the members.
- Travel angle: inclination along the direction of welding. A slight drag or push angle is commonly used depending on process and position.
- Incorrect angle can cause undercut, unequal leg size, slag entrapment, lack of fusion or excessive overlap.

7. Carbon Arc Welding and Shielded Arc Welding

7.1 Carbon arc welding

Carbon arc welding uses an arc between a carbon/graphite electrode and the workpiece. The carbon electrode is essentially non-consumable; filler metal, if required, is supplied separately. Single-carbon and twin-carbon arrangements are possible.

- Advantages: simple equipment, low electrode cost, useful for heating, brazing and some non-ferrous work.
- Limitations: risk of carbon pickup in ferrous welds, relatively poor shielding, lower productivity than modern gas-shielded processes.

7.2 Shielded metal arc welding (SMAW)

In shielded metal arc welding, the arc is struck between a flux-coated consumable electrode and the work. The coating decomposes to provide shielding gas and forms a slag layer over the molten weld metal. The slag protects the hot weld from atmospheric contamination and is removed after solidification.

- Portable and versatile; can weld in many positions and outdoor conditions.
- Suitable for construction, maintenance, pipelines, structural fabrication and repair.
- Limitations include frequent electrode changing, slag removal, fumes/spatter and lower deposition rate than continuous-wire processes.

8. Important Points

- Weldability depends on material, process, joint design, heat input and service requirement.
- Resistance welding heat is approximately proportional to I^2Rt, with pressure applied at the joint.
- Neutral, carburizing and oxidizing are the three principal oxy-acetylene flame types.
- AC welding minimizes arc blow; DC permits deliberate polarity selection.
- Current mainly governs melting rate and penetration; arc voltage is strongly influenced by arc length.
- SMAW uses a flux-coated consumable electrode that provides shielding and slag.